

Khayra Sarra Zerguerras

Final-Year AI & Data Science Engineering Student at ESI-SBA | Seeking Internship – January 2027

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Summary

Final-year AI & Data Science Engineering student at ESI-SBA with hands-on experience in machine learning, deep learning, NLP, generative AI, data engineering and MLOps. Experienced in developing end-to-end AI and data applications from model development and experimentation to deployment, with additional full-stack development skills. Seeking an AI, Data Science or Data Engineering internship starting January 2027.

Technical Skills

Programming: Python, SQL, Java, JavaScript, TypeScript, Dart, PL/SQL

AI & ML: Machine Learning, Deep Learning, NLP, Generative AI, Conditional GANs, Knowledge Engineering, MLOps

Data Engineering: Engi-PySpark, Apache Spark (RDD, Spark SQL, MLlib), Hadoop, Apache Airflow, ETL, Data Lakes, Stream Processing, GraphFrames

Data Databases: &PostgreSQL, MySQL, MongoDB, NoSQL, Data Warehousing, SQL OLAP, PCA, Data Analysis, Power BI

Development: Streamlit, React, Next.js, Flutter, Django, Node.js, Express.js, Spring Boot, HTML5, CSS3, Tailwind CSS

Cloud & Tools: AWS (EC2, S3), Docker, Git, Postman, Figma

Selected Projects

Sign Language Production – Text-to-Video

- Developed an end-to-end text-to-ASL generation pipeline using 3D skeleton poses and diffusion models.
- Designed transformer-based text conditioning with cross-attention and contrastive alignment, eliminating the need for an intermediate gloss representation.

Smart Notes Generator – Audio-to-Impact

- Developed an AI application transforming raw audio into structured notes and concise summaries using NLP and language models.

Market Basket Analysis & Product Recommendation

- Built a product recommendation system using association rule mining and implemented an MLOps workflow for reproducible model development and deployment.

AI-Powered Job Skills Gap Analyzer

- Developed a Streamlit ML application analyzing gaps between user skills and job requirements to provide data-driven insights.

Automatic Image Colorization

- Implemented a Conditional Generative Adversarial Network (cGAN) for automatic grayscale image colorization.

Malicious URL Detector

- Built and compared classification algorithms for detecting phishing, malware, spam and defacement URLs.

Professional Experience

CNR (Caisse Nationale des Retraites) – Chlef

Summer 2025

Intern – Software Development Department

- Completed a 4th-year engineering internship focused on internal software and data systems; analyzed internal/external applications and documented optimization recommendations.

Alphabit Club

2024–Present

Finance Department

- Managed event budgets and prepared financial reports for club activities.

Ingenuims Club

2022–2024

Marketing / Visual Creation Department

- Created visual content and photography for student events and activities.

Education

École Supérieure en Informatique (ESI-SBA)

2022–2027

Engineering Degree in AI & Data Science – Expected 2027 | Final year of engineering cycle

Baccalaureate in Mathematics

2022 | Graduated with honors – 17.36/20

Certifications

Oracle Certified Generative AI Professional  | AWS Concepts  | Cloud Computing 

Languages

Arabic (Native) | French (Fluent) | [English \(C1 Advanced – EF SET, 63/100\)](#)